

Adventist University of Central Africa

Faculty of Information Technology

INSY 8311: Database Development with PL/SQL (Group B)

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Project Proposal: Airport Operations Management System

November 13, 2026

Airport Operations Management System

Problem Statement

Technological advances in aviation are outpacing small airports' ability to adapt. These airports often lack specialized IT staff and cannot afford robust IT solutions (Airport Cooperative Research Program, 2015). As a result, they struggle to meet their operational needs within budget constraints.

Proposed Solution

For this reason, I propose a cost-free, easier-to-manage solution: Airport Operations Management System (AOMS). This system navigates around the specialized IT staff issue since it is largely automated and requires minimal tech skills. Additionally, small airports can adopt it without making financial investments.

Objective

The AOMS aims to manage airport-side operational information, such as staff, runways, equipment, ground operations, inspection, and air traffic control, through a normalized database that integrates with Power BI to display data visually with accurate and meaningful reports.

Data Source

My primary data source is from Kaggle, and it was uploaded by Shereef (n.d.). It simulates an airport operations database used at Indira Gandhi International Airport. Since it offers incomplete tables, other real-world datasets will be adapted to complete my planned database.

Planned Tables

Table	Description
Staff	Stores information about staff at the airport
ATC	Stores additional information on air traffic controllers (Specialized staff)
Equipment	Stores information on airport equipment such as aircraft, baggage handling equipment, light systems, ground support equipment, and emergency equipment.
Runways	Stores information about runways and taxiways at the airport.
Maintenance logs	Logs of ground operations on aircraft as well as equipment maintenance
Inspection	Inspection logs of airport equipment

Planned PL/SQL Features

My project will implement the following PL/SQL features

Feature	Description
Transactions	Logical units of work that execute either all-or-nothing operations on the database.
Procedures	No-return subroutines that perform specific tasks
Functions	Subprograms that manipulate data and return values
Exception handling	Mechanism that allows to catch errors related to invalid database operations or invalid user input
Records & Collections	Composite data types that hold multiple elements together for efficient data handling
Cursor	Pointer to query results that allows for row-by-row processing and advanced query processing logic
Trigger	Programs that automatically execute when some events occur, accommodating audits and event logging.
Package	Schema objects that group related items, such as variables, data types, and subprograms into one unit for modularity.

Innovation

The technology I chose to add an innovation layer to my project is Microsoft Power BI. I will build an analytics dashboard to summarize key metrics in the database and produce clean, accurate reports to support decision-making.

References

Airport Cooperative Research Program. (2015). Alternative IT delivery methods and best practices for small airports. *The National Academies Press*. https://doi.org/10.17226/22198

Shereef. (n.d.). *Airport operations multi-table dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/sinanshereef/airport-operations-multi-table-dataset